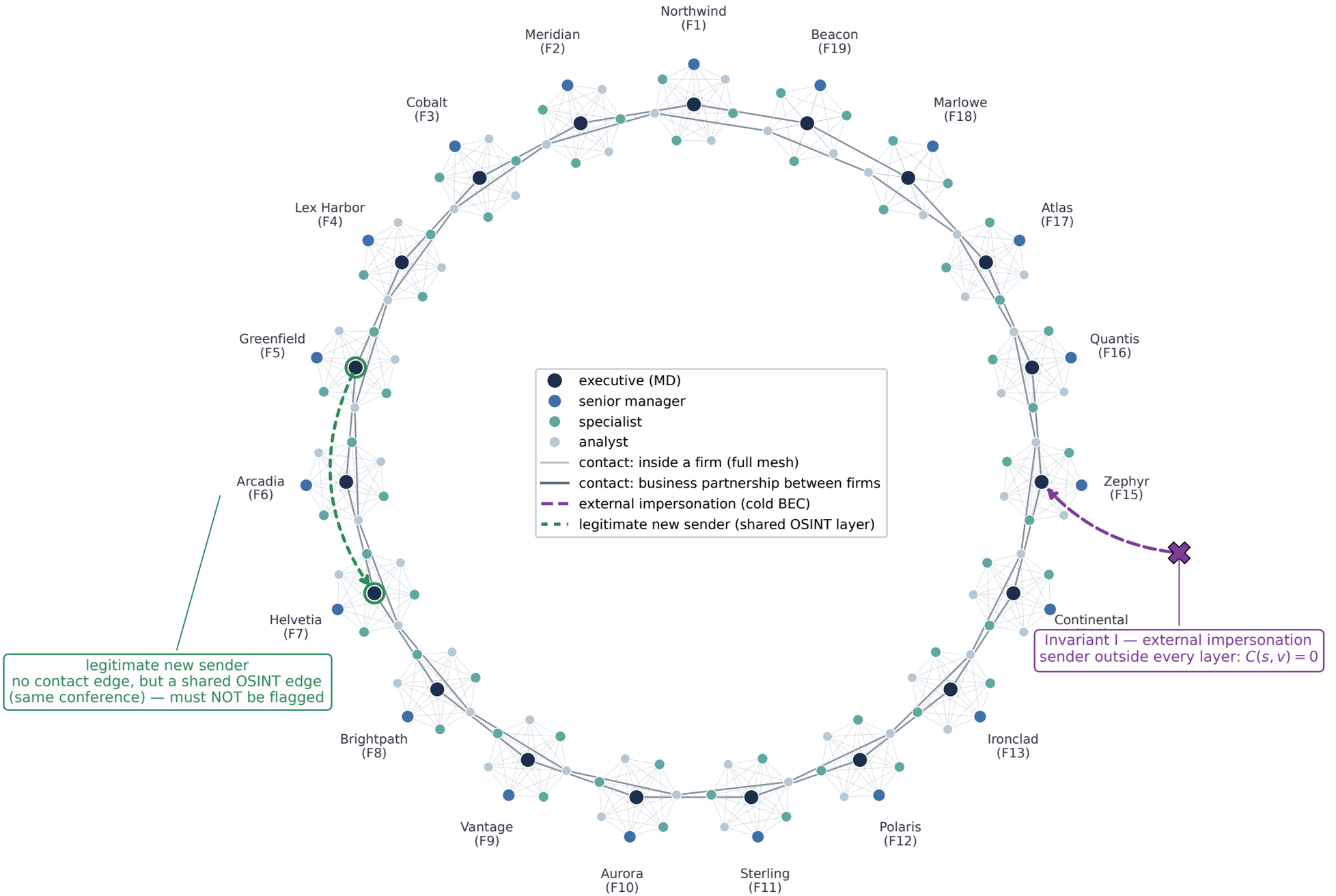
Case-study organization: 19 firms, 150 accounts, 556 contact edges (518 inside firms, 38 between partner firms)
Northwind (F1)
Meridian (F2)
Beacon (F19)
Cobalt (F3)
Marlowe (F18)
Lex Harbor (F4)
Atlas (F17)
Greenfield (F5)
Quantis (F16)
Arcadia (F6)
Zephyr (F15)
Helvetia (F7)
Continental
Brightpath (F8)
Ironclad (F13)
Vantage (F9)
Polaris (F12)
Aurora (F10)
Sterling (F11)
executive (MD)
senior manager
specialist
analyst
contact: inside a firm (full mesh)
contact: business partnership between firms
external impersonation (cold BEC)
legitimate new sender (shared OSINT layer)
legitimate new sender
no contact edge, but a shared OSINT edge
(same conference) — must NOT be flagged
Invariant I — external impersonation
sender outside every layer: $C(s, v) = 0$